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Insurance & Econometrics

Final Project - Solvency II

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0 Original text of the Project

Consider a simplified insurance company whose assets and liabilities sides are characterized as follows:

ASSETS

- For each policyholder there is a single fund made of equity (80%) and property (20%):

$$F_t = EQ_t + PR_t$$

- At the beginning ($t = 0$) the value of the fund is equal to the invested premium:

$$F_0 = C_0 = 100,000$$

- **Equity features:**

- Listed in the regulated markets in the EEA.
- No dividend yield.
- To be simulated with a Risk Neutral GBM with $\sigma = 20\%$ and a time varying forward rate.

- **Property features:**

- Listed in the regulated markets in the EEA.
- No dividend yield.
- To be simulated with a Risk Neutral GBM with $\sigma = 10\%$ and a time varying forward rate.

LIABILITIES

- **Contract terms:**

- Whole Life policy.

- **Benefits:**

- * In case of *lapse*, the beneficiary gets the value of the fund at the time of lapse, with 20€ of penalties applied.
- * In case of *death*, the beneficiary gets the maximum between the invested premium and the value of the fund: $\max(C_0, F_t)$.

- **Others:**

- * Regular Deduction (RD) of 2.20%.
- * Commissions to the distribution channels, COMM (or trailing) of 1.40%.

- **Model points:**

- Just 1 model point of 100 insured.
- Each insured is a male with insured aged $x = 60$ at the beginning of the contract.

- **Operating assumptions:**

- Mortality: rates derived from the life table SI2024.
- Lapse: flat annual rates $l_t = 15\%$.
- Expenses: constant unitary (i.e. per policy) cost of 50€ per year, that grows following the inflation pattern.

- **Economic assumption:**

- Risk free: rate r derived from the yield curve (EIOPA EUR without VA at 31.12.25).
- Inflation: flat annual rate of 2%.

OTHER SPECIFICATIONS

- Time step: annual.
- Time horizon for the projection: 50 years.
- In case of outstanding portfolio in $T = 50$, let all the people leave the contract with a massive surrender (no penalties applied).
- The interest rates dynamic is deterministic, while the equity and property ones are stochastic.
- Apply the symmetric adjustment to the equity risk charge.

1 Summary Results

1.1 Stochastic Table

Results	MVA	BEL	Δ MVA abs	Δ BEL abs	Δ MVA rel	Δ BEL rel	BoF	Δ BoF	dur_L
BASE	10,000,000.00	9,657,338.26	-	-	-	-	342,661.74	-	5.54
IR_UP	10,000,000.00	9,630,276.82	0	-27,061.44	0%	-0.28%	369,723.18	0	-
IR_DOWN	10,000,000.00	9,687,680.76	0	30,342.50	0%	0.31%	312,319.24	30,342.50	-
EQUITY	6,248,000.00	6,199,472.88	3,752,000.00	-3,457,865.38	37.52%	-35.81%	48,527.12	294,134.62	-
PROPERTY	9,500,000.00	9,196,788.90	500,000.00	-460,549.36	5%	-4.77%	303,211.10	39,450.64	-
MORTALITY	10,000,000.00	9,670,255.24	0	12,916.98	0%	0.13%	329,744.40	12,916.98	-
EXPENSES	10,000,000.00	9,662,255.60	0	4,917.34	0%	0.05%	337,744.40	4,917.34	-
CATASTROPHY	10,000,000.00	9,658,704.70	0	1,366.44	0%	0.01%	341,295.30	1,366.44	-
LAPSE_UP	10,000,000.00	9,737,471.02	0	80,132.76	0%	0.83%	262,528.98	80,132.76	-
LAPSE_DOWN	10,000,000.00	9,546,457.96	0	-110,880.31	0%	-1.15%	453,542.04	0	-
LAPSE_MASS	10,000,000.00	9,765,284.23	0	107,945.97	0%	1.12%	234,715.77	107,945.97	-
BSCR								384,825.52	

Table 1: Stochastic projections¹

1.2 Deterministic Table

Results	MVA	BEL	Δ MVA abs	Δ BEL abs	Δ MVA rel	Δ BEL rel	BoF	Δ BoF	dur_L
BASE	10,000,000.00	9,577,794.21	-	-	-	-	422,205.79	-	5.64
IR_UP	10,000,000.00	9,575,773.79	0	-2,020.41	0%	-0.0211%	424,226.21	0	-
IR_DOWN	10,000,000.00	9,594,005.00	0	16,210.79	0%	0.1693%	405,995.00	16,210.79	-
EQUITY	6,248,000.00	6,179,444.84	3,752,000.00	-3,398,349.37	37.52%	-35.4815%	68,555.16	353,650.63	-
PROPERTY	9,500,000.00	9,113,829.90	500,000.00	-463,964.30	5.00%	-4.8442%	386,170.10	36,035.70	-
MORTALITY	10,000,000.00	9,580,638.37	0	2,844.16	0%	0.0284%	419,361.63	2,844.16	-
EXPENSES	10,000,000.00	9,582,711.54	0	4,917.34	0%	0.0513%	417,288.46	4,917.34	-
CATASTROPHY	10,000,000.00	9,578,343.17	0	548.96	0%	0.0057%	421,656.83	548.96	-
LAPSE_UP	10,000,000.00	9,697,430.29	0	119,636.08	0%	1.2491%	302,569.71	119,636.08	-
LAPSE_DOWN	10,000,000.00	9,322,844.52	0	-254,949.69	0%	-2.6619%	677,155.48	0	-
LAPSE_MASS	10,000,000.00	9,715,934.93	0	138,140.72	0%	1.4423%	284,065.07	138,140.72	-
BSCR								446,508.25	

Table 2: Deterministic projections²

²¹ To ensure a consistent interpretation across all reported results, the variations (Δ) are defined such that positive values represent an adverse impact on the company's financial position. Specifically, the capital requirement is determined as $\Delta\text{BoF} = \max(0, \text{Base} - \text{Shock})$, following the Solvency II principle where the loss is floored at zero. Accordingly, to align with the "positive-as-loss" framework, asset and liability fluctuations are expressed as $(\text{Base} - \text{Shock})$ for MVA and $(\text{Shock} - \text{Base})$ for BEL.

2 Formula Specifications

2.1 Technical Modeling Specifications

The valuation of the contract begins with the collection of the single premium C_0 an instant before the start of the first year of projection. This timing convention ensures that at $t = 0$, the insurance coverage and the investment in the internal fund formally starts with the full premium amount, while the policyholders are aged $x = 60$.

Regarding the settlement of liabilities, all outflows are recognized at the end of each year of projection. If a policyholder dies during year t , the benefit paid at year-end is defined as the maximum between the invested premium and the fund value, considered net of the annual Regular Deduction (RD). Similarly, in the event of a lapse occurring during year t , the payout corresponds to the fund value at the end of the year (net of RD), minus the applicable penalty.

Operational costs, specifically expenses and trailing distribution commissions, are sustained for all policyholders who are alive and have not lapsed at the beginning of year t . Although these costs are determined by the status of the policyholders at the beginning of the period, the relative cash flows are accounted for at the end of the projection year.

2.2 Asset Modeling and Scenario Generation

The portfolio consists of two main assets: Equity (80%) and Property (20%). We model both using Geometric Brownian Motion (GBM), which is the standard method for Solvency II calculations.

2.2.1 The Stochastic Model

The growth of each asset is described by the following formula:

$$S_t = S_{t-1} \cdot \exp \left(\left(f_t - \frac{1}{2}\sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_t \right)$$

Where:

- f_t is the compounded forward rate in $(t, t + \Delta t)$
- σ is the constant volatility: 20% for Equity and 10% for Property.
- $Z_t \sim \mathcal{N}(0, 1)$ is the random market shock.

2.2.2 Simulation Method: Antithetic Variates

To make the simulation more efficient and reduce "noise", we use the **Antithetic Variates** technique. We generate random shocks for only half of the simulations ($N/2$). For the other half, we use the exact negative of those shocks.

This ensures that the average shock in our model is exactly zero. By doing this, we "sterilize" the statistical error. This means that when we compare the base case with the stressed cases, the difference (ΔBOF) is caused only by the specific shock we are testing, and not by random chance in the Monte Carlo paths.

2.2.3 Model Validation

We validated our model using the **Martingale Test**. According to financial theory, the discounted price of an asset should be a martingale. This means its average discounted value in the future must equal its value today:

$$\mathbb{E}^{\mathbb{Q}} [D(0, t)S_t] = S_0$$

We chose to run ****1 million simulations**** ($N = 10^6$) for the following reasons:

- **Equity Volatility:** Since Equity has a high volatility (20%), it creates a lot of variation. We need a very large number of scenarios to make sure the average result is stable and reliable.
- **Accuracy:** With 1 million scenarios, the Martingale Test for Equity showed an error of less than 0.26% . This precision is necessary for calculating the PVFP correctly over a 50-year horizon.
- **Stability:** A smaller number of scenarios (like 10,000) would leave too much "noise" in the results, making the SCR calculation less precise. Using 10^6 paths ensures that our results are robust and repeatable.

2.3 Liabilities

Cash flows are discounted using the **EIOPA EUR yield curve** (without Volatility Adjustment) as of 31.12.2025. The discount factor for each year of projection t is defined as:

$$DF_t = \frac{1}{(1 + r_t)^t}$$

where r_t represents the annually compounded spot rate for maturity t .

The portfolio evolution is governed by two independent decrement causes. Mortality rates (q_x) are derived from the **ISTAT SI2024** life table for the male population. Lapses are modeled with a flat annual rate of 15%, with a mandatory mass surrender applied at the end of the 50th year of projection to clear the outstanding portfolio without further penalties.

The active population at the beginning of period t , denoted L_t , evolves recursively as:

$$L_1 = N, \quad L_{t+1} = L_t - \text{Deaths}_t - \text{Lapses}_t$$

where $N = 100$ is the initial number of policyholders. The two decrement causes are applied sequentially within each period. Deaths occur first:

$$\text{Deaths}_t = L_t \cdot q_{x+t-1}$$

Lapses are then applied to the surviving population:

$$\text{Lapses}_t = \begin{cases} (L_t - \text{Deaths}_t) \cdot l & t < 50 \\ L_t - \text{Deaths}_t & t = 50 \end{cases}$$

where $l = 15\%$ is the flat annual lapse rate. At $t = 50$, all remaining policyholders surrender without penalty, clearing the outstanding portfolio.

Before describing the BEL components, we define the two fund values appearing in the cash flow formulas. In the **deterministic projection** (Excel), the fund grows period by period by a factor of $(1 + f_t)$, where f_t is the annually compounded forward rate derived from the EIOPA curve, with no volatility. In the **stochastic projection** (Python), the fund dynamics follow a Risk-Neutral GBM, as described in Section 2.2.1. In both cases, the Regular Deduction is applied to obtain the net fund value:

$$F_{t,\text{afterRD}} = F_{t,\text{beforeRD}} \cdot (1 - RD)$$

where $RD = 2.20\%$. $F_{t,\text{afterRD}}$ is carried forward to the next period and used to compute death and lapse benefits, while $F_{t,\text{beforeRD}}$ serves as the base for commission calculations. The initial fund value is $F_{0,\text{afterRD}} = C_0 = 100,000\text{€}$ per policyholder.

The total BEL is decomposed into four primary components, consistent with the timing conventions described in Section 2.1:

1. Death Liabilities:

$$BEL_{\text{Death}} = \sum_{t=1}^{50} \text{Deaths}_t \cdot \max(C_0, F_{t,\text{afterRD}}) \cdot DF_t$$

2. Lapse Liabilities:

$$BEL_{\text{Lapse}} = \sum_{t=1}^{50} \text{Lapses}_t \cdot \max(0, F_{t,\text{afterRD}} - P_t) \cdot DF_t$$

where $P_t = 20\text{€}$ for $t < 50$ and $P_t = 0$ at $t = 50$.

3. Expense Liabilities:

$$BEL_{\text{Exp}} = \sum_{t=1}^{50} L_t \cdot \text{Expense}_0 \cdot (1 + \pi)^t \cdot DF_t$$

where $\text{Expense}_0 = 50\text{€}$ and $\pi = 2\%$.

4. Commission Liabilities:

$$BEL_{\text{Comm}} = \sum_{t=1}^{50} L_t \cdot \text{COMM} \cdot F_{t,\text{beforeRD}} \cdot DF_t$$

The final Best Estimate of Liabilities is obtained by summing the four components:

$$BEL = BEL_{\text{Death}} + BEL_{\text{Lapse}} + BEL_{\text{Exp}} + BEL_{\text{Comm}}$$

3 BSCR Aggregation Framework

The BSCR is computed by aggregating individual risks from two broad categories — Market and Life — through linear correlations, which account for diversification and mitigation effects.

Market Risk captures the exposure to fluctuations in the level and volatility of financial instruments held in the portfolio, specifically changes in equity and property prices as well as interest rates. Life Underwriting Risk, on the other hand, consolidates the capital requirements for mortality, lapse, expense, and CAT risks. Each category is first aggregated internally using the respective correlation matrix:

$$Corr_{mkt} = \begin{matrix} & \text{Int} & \text{Eq} & \text{Prop} \\ \begin{matrix} \text{Int} \\ \text{Eq} \\ \text{Prop} \end{matrix} & \begin{bmatrix} 1 & A & A \\ A & 1 & 0.75 \\ A & 0.75 & 1 \end{bmatrix} \end{matrix} \quad Corr_{life} = \begin{matrix} & \text{Mort} & \text{Lap} & \text{Exp} & \text{CAT} \\ \begin{matrix} \text{Mort} \\ \text{Lap} \\ \text{Exp} \\ \text{CAT} \end{matrix} & \begin{bmatrix} 1 & 0 & 0.25 & 0.25 \\ 0 & 1 & 0.5 & 0.25 \\ 0.25 & 0.5 & 1 & 0.25 \\ 0.25 & 0.25 & 0.25 & 1 \end{bmatrix} \end{matrix}$$

The intra-class Solvency Capital Requirement is then obtained as:

$$SCR = \sqrt{\sum_{ij} Corr_{i,j} SCR_i SCR_j}$$

The value of the parameter A depends on the behavior of the portfolio. If the insurance company is exposed to an increase in interest rates, then A= 0. Otherwise, if the insurance company is exposed to a decrease in interest rates, then A= 0.5.

Once both intra-class requirements have been computed, the Market and Life components are combined into the vector $\mathbf{SCR} = [SCR_{mkt}, SCR_{life}]$ and aggregated through a final inter-class correlation structure, yielding the BSCR:

$$BSCR = \sqrt{\mathbf{SCR}^T C \mathbf{SCR}}, \quad C = \begin{bmatrix} 1 & 0.25 \\ 0.25 & 1 \end{bmatrix}$$

Throughout the following subsections, all numerical results refer to the stochastic projection. As discussed in Section 4.1, the direction and sign of each shock are consistent across both approaches, making the stochastic figures representative of the overall model behavior.

3.1 Interest Rates

To evaluate the portfolio's sensitivity to interest rate fluctuations, we applied the EIOPA upward (IR_{up}) and downward (IR_{down}) shock scenarios. A change in interest rates has a dual effect on the balance sheet: it modifies the expected accumulation of the fund while also shifting the discounted value of future liabilities.

$$\begin{aligned} MkT^{up} &= \Delta BOF \mid IR_{up} \\ MkT^{down} &= \Delta BOF \mid IR_{down} \end{aligned}$$

To determine the final capital charge for this risk module, we select the most adverse outcome between the two simulated scenarios:

$$MkT_{ir} = \max \left\{ MkT^{up}, MkT^{down} \right\}$$

As our results indicate, the IR_{down} scenario acts as the binding constraint, setting the definitive capital requirement for interest rate risk at approximately 30,000.

3.2 Market Equity

To quantify the portfolio's exposure to sudden stock market downturns, we stressed the equity components by applying a base 39% decline, coupled with the updated EIOPA symmetric adjustment of 7.89%.

$$SCR_{eq} = \Delta BOF \mid \text{equity shock}$$

Given that equities dominate the initial asset allocation at 80%, this combined stress triggers a severe contraction in the Market Value of Assets (MVA). From an economic perspective, this asset collapse is partially mitigated by a significant reduction in the Best Estimate of Liabilities (BEL). This occurs because the contract is unit-linked: as the fund value drops, future lapse and death benefit payments decrease accordingly, since both are directly tied to the fund value. However, because the capital loss on the asset side far outweighs this "liability relief," the shock results in a substantial net depletion of the Basic Own Funds (BOF), confirming that equity volatility remains the primary market risk for the company.

3.3 Market Property

To assess the portfolio's vulnerability to real estate downturns, we subjected the property holdings to a 25% depreciation shock, without applying any symmetric adjustment mechanism.

$$SCR_{pr} = \Delta BOF \mid \text{property shock}$$

Since real estate constitutes 20% of the initial asset allocation, the 25% localized shock translates to a modest 5% decline in the total Market Value of Assets.

3.4 Mortality

To evaluate the portfolio's exposure to adverse biometric trends, we applied an immediate and permanent 15% upward shock to the base mortality rates.

$$SCR_{mort} = \Delta BOF \mid \text{mortality shock}$$

Unlike financial shocks, this is a pure underwriting risk that leaves the investment portfolio entirely unaffected. The stress manifests strictly as a liability increase, translating directly into an equivalent loss of capital. This impact is strikingly low due to the portfolio's demographic dynamics: early policy surrenders (lapses) drive the main outflows. By the time natural mortality becomes statistically significant at older ages, the majority of policyholders have already exited the fund, naturally hedging the company's mortality exposure.

3.5 Lapses

Lapse risk captures the sensitivity of the BEL to unexpected changes in policyholder behavior. The Standard Formula prescribes three stress scenarios, with markedly different effects on the balance sheet.

Upward: Baseline lapse rates are permanently increased to 22.5%, according to the formula $\min\{100\%, 150\% \cdot R\}$. Accelerated exits mean the insurer loses years of expected fee collections. This drops the portfolio's future profitability, raising technical liabilities by roughly 80,000 and depleting capital by the exact same margin.

$$SCR_L^{UP} = \Delta BOF \mid \text{lapse shock up}$$

Downward: Rates are heavily reduced to 7.5%, according to the formula provided by the authority $\max\{50\% \cdot R, R - 20\%\}$. Counterintuitively for a "shock", a decrease in lapses fundamentally benefits the company: policyholders stay longer, generating more prolonged Regular Deductions. This scenario actually yields a capital gain, zeroing out the capital requirement (SCR) for this specific leg.

$$SCR_L^{DOWN} = \Delta BOF \mid \text{lapse shock down}$$

Instantaneous Mass Lapse Shock: In alignment with the Solvency II Standard Formula, the mass lapse shock is modeled as an instantaneous, one-off event occurring at the beginning of the projection year ($T=0$). This stress test mandates an immediate lapse of 40% of the total in-force portfolio, a shock applied over and above the regular best estimate lapse assumptions. Consequently, for the remainder of the year, the standard best estimate lapse rates are applied only to the surviving 60% of the portfolio, ensuring the liability calculation captures both the extreme event and the underlying policyholder behavior within the stressed scenario.

$$SCR_L^{MASS} = \Delta BOF \mid \text{lapse mass shock}$$

To establish the final Solvency Capital Requirement for lapse risk, regulatory frameworks dictate a strict prudence principle. We must select the most punishing of these three simulated outcomes to ensure the company holds a robust margin of safety against unpredictable policyholder behavior:

$$SCR_L = \max \left\{ SCR_L^{UP}, SCR_L^{DOWN}, SCR_L^{MASS} \right\}$$

As clearly reflected in our numerical results, the Massive Surrender event represents the absolute worst-case scenario for this specific portfolio. While a downward shock actually strengthens the balance sheet by prolonging fee collection, a sudden 40% mass exodus instantaneously wipes out a massive portion of future Regular Deductions before they can be harvested. Consequently, this mass surrender scenario drives the highest liability increase and determines the definitive capital charge of approximately 108,000 euros for the lapse risk module.

3.6 Life Cat

The Life Catastrophe risk module is designed to account for extreme, tail-end events—such as a global pandemic—that standard mortality stresses fail to capture. Following regulatory requirements, we modeled this scenario by applying a flat 0.15% increase in the mortality rate specifically for the first year of the contract.

$$SCR_{cat} = \Delta BOF \mid \text{CAT}$$

As shown in our results, the financial impact is minimal, leading to a Basic Own Funds depletion of only 1,366.44. This effect is even more contained than the standard mortality shock because the surge in deaths is temporary rather than permanent.

3.7 Expenses

To account for operational and macroeconomic volatility, we stressed the initial maintenance cost of €50 per policy and the 2% base inflation rate. This risk module captures potential surges in administrative overhead, such as rising labor costs or unexpected inflationary pressures, which typically correlate over long horizons.

$$SCR_{exp} = \Delta BOF \mid \text{expense shock}$$

We modeled this scenario by applying a 10% increase to the unitary expense alongside a 1% flat hike in the inflation rate. In our portfolio, this shock leads to a capital depletion of 4,917.34. While the impact is strictly negative as it increases the Best Estimate of Liabilities, the overall effect on solvency is limited due to the relatively low weight of management costs compared to the total fund value.

4 Deterministic Projections

In this section we replicate the previous calculations under a deterministic framework. All assumptions are kept unchanged: the only difference with respect to the stochastic projection is that equity and property grow at the time-varying forward rate with no volatility.

4.1 Comparison with Stochastic Projections

We begin by comparing the base case results. The BEL is higher in the stochastic case, and decomposing by component, the gap is almost entirely driven by the death benefit: lapse, expenses and commissions are nearly identical across the two approaches. This is explained by the guarantee $\max(C_0, F)$ embedded in the death benefit: in the deterministic projection the fund grows monotonically after the first two years and never falls below C_0 , so the guarantee rarely bites. In the stochastic projection the fund can fall below C_0 along adverse paths, making the guarantee more valuable and increasing the corresponding BEL component.

Moving to the shocked scenarios, the two approaches yield consistent results in terms of direction and sign, confirming the robustness of the model: scenarios that negatively impact the company are coherent across both methods. Risks involving the death benefit guarantee, such as $SCR_{Mortality}$ and SCR_{CAT} , are systematically higher in the stochastic case, where adverse market paths increase the probability of the guarantee being activated. Conversely, lapse-related SCRs are higher in the deterministic case, as the monotonic growth of the fund results in higher lapse benefit payments compared to the stochastic average. $SCR_{Expenses}$ is identical in both cases, as expected, since the expense shock does not interact with the fund dynamics.

4.2 Macaulay Duration of Liabilities

The Macaulay Duration (MacD) represents the weighted average time until the policyholder liabilities are paid.

$$MacD = \frac{\sum_{t=1}^T t \cdot BEL_t}{\text{Total Liabilities}}$$

For our base case, the portfolio exhibits a relatively short duration due to the high lapse assumptions (which naturally accelerate the outflows), yielding:

$$MacD = 5.64 \text{ years}$$

4.3 PVFP

The PVFP quantifies the expected net value generated by the policy over its lifetime, discounted to the present. The source of revenue for the insurer is the Regular Deduction (RD), applied at the end of each year to the gross fund value before being returned to the policyholder. The sources of cost are commissions paid to distribution channels and fixed expenses. The annual profit is therefore:

$$\Pi_t = RD_t - \text{Commissions}_t - \text{Expenses}_t$$

where $RD_t = RD \cdot F_{t,\text{beforeRD}} \cdot L_t$ is the total Regular Deduction collected across all active policyholders at time t . The total profit stream is then discounted:

$$PVFP = \sum_{t=1}^{50} \Pi_t \cdot DF_t$$

Obtaining:

$$PVFP = 420,794.49 \text{ €}$$

The PVFP represents approximately 4.2% of the initial MVA, reflecting the thin margin structure of this product. The insurer retains only $RD - COMM = 0.80\%$ of the fund as net fee margin, which is a notably low spread. Fixed expenses further compress this margin, confirming that the product leaves limited space for profit generation.

4.4 Leakage

The leakage is the residual difference in the valuation equation:

$$MVA = BEL + PVFP + \text{Leakage}$$

$$\text{Leakage} = 10,000,000.00 - 9,577,794.21 - 420,794.49 = 1,411.31 \text{ €}$$

Its magnitude of approximately 0.014% of the initial MVA confirms that the model is consistent and that the approximation error is negligible.

4.5 PVFP via Rule of Thumb

As a sense check, we verify the PVFP by approximating it as the product of the first-year profit and the Macaulay Duration of the liabilities:

$$PVFP \approx DU_{\text{Macaulay}} \cdot PVFP(1)$$

The results are:

Method	Value
Proxy (Rule of Thumb)	423,248.11 €
Exact PVFP	420,794.49 €

The two values are in close agreement, with a relative difference of approximately 0.58%, confirming the consistency of our PVFP calculation.

5 Open questions

5.1 What happens to the asset and liabilities when the risk-free rate increases/decreases with a parallel shift of, say, 100bps?

The parallel shift of the interest rate (+/- 100bps) impacts two of the main components in our analysis: the **forward growth rate** of the fund and also the **discount factor** which we use to discount the future cash flows.

The MVA at $t=0$ does not change since it consists of a single premium given at the start, so we are not shocked by the fact that it does not change when the risk-free rates increase or decrease by 100bps. However, what we expect to change are the various components of the BEL and they change differently with respect to the type of component we are studying

1. **Expenses:** Expenses do not depend on the fund's value evolution. Since they are deterministic cash flows, they are only impacted by the discounting effect. In particular, an increase in rates, increases the discount factors and so decreases the present value of the expenses. Conversely, a decrease in rates leads to an increase in the expenses.
2. **Commissions and Surrenders:** One might expect that if the rates go up, so the fund value goes up as well, then the commissions will go up as well. This is of course true, but we have to remember also the fact that also the discount factors go up as well. In a risk-neutral setting, the discounted fund value follows a **martingale property**:

$$E[\text{Fund}(t)DF_t] = \text{Fund}(0) \quad (1)$$

Since the commissions are calculated as a proportional fee of the fund, the two main components presented at the start of the discussion (growth of the fund and the discount factor) perfectly offset each other, leaving the Commission part of the BEL identical across all scenarios. The same type of analysis can be applied also to the Surrender part, where the small variations are due to the fixed penalty, but we think they are quite negligible.

3. **Death benefits:** the Death part of the BEL changes quite significantly in the different scenario that we studied. As presented in the discussion on the formulas, we know that the payout is not directly proportional to the value of the fund, but it includes the following guarantee:

$$\text{Payout} = \max(F_0, \text{Fund}_{\text{afterRD}}(t)) \quad (2)$$

We know that if the rates go down, the fund grows at a slower pace, increasing the probability that its value will go below F_0 . So the intrinsic value becomes more expensive, raising the death liability. On the opposite, if rates go up, the fund grows its value, so the payout will be the value of the fund most of the time, reducing the cost of the guarantee.

BEL Component	Deterministic (€)	Stochastic (€)
Death	582,203	662,727
Surrender	8,176,810	8,175,919
Expenses	29,959	29,960
Commissions	788,820	788,732
BASE TOTAL	9,577,794	9,657,338

Table 3: Breakdown of the BEL (Deterministic vs. Stochastic) in the Base Case.

Scenario	BEL Component	Deterministic (€)	Stochastic (€)
+100 bps	Death	582,018	640,486
	Surrender	8,176,887	8,175,997
	Expenses	28,383	28,383
	Commissions	788,820	788,732
	UP TOTAL	9,576,110	9,633,599
-100 bps	Death	597,628	692,875
	Surrender	8,176,724	8,175,834
	Expenses	31,710	31,711
	Commissions	788,820	788,732
	DOWN TOTAL	9,594,884	9,689,152

Table 4: Breakdown and sensitivity analysis of the BEL (Deterministic vs. Stochastic) to interest rate variation (+/- 100bps)

5.2 What happens to the liabilities if the insured age increases? What if there were two model points, one with males and one with females?

Let's first consider the scenario in which the insured age increases. We assume the policyholders enter at age 65.

Base Case

Components	Deterministic	Stochastic
Expenses	29,959€	29,960€
Commissions	788,820€	788,732€
Death	582,203€	662,727€
Lapses	8,176,810€	8,175,919€
BEL	9,577,794€	9,657,338€

PH enters the contract aged 65

Components	Deterministic	Stochastic
Expenses	29,040€	29,041€
Commissions	766,343€	766,271€
Death	900,955€	1,024,250€
Lapses	7,893,544€	7,892,824€
BEL	9,589,884€	9,712,385€

In this case, we observe a clear increase in the Best Estimate Liability (BEL), primarily driven by a significant rise in death benefit liabilities. This is expected, as mortality rates from the SI2024 life table increase sharply with age beyond 60. Compared to the base case, an older policyholder is more likely to trigger the death benefit earlier, which not only reduces discounting but also increases the probability of the guaranteed amount (100,000 €) being in-the-money relative to the fund. This leads to a higher present value of the death benefit.

This behavior is somewhat analogous to applying a mortality shock in the Solvency

II framework. However, there's a subtle difference: a mortality shock increases all death probabilities by a fixed percentage, while increasing the starting age changes the structure of the mortality curve itself, with higher relative growth in early years and more varied dynamics over time.

Now let's consider the case of having two model points: one for males and one for females, each representing a portfolio of 100 policyholders aged 60 at inception, with an individual initial premium of 100,000€, for a total MVA of 10,000,000€.

Scenario	Stochastic BEL	Deterministic BEL
Male	9,657,338 €	9,577,794 €
Female	9,619,677 €	9,569,516 €
Δ (Male – Female)	37,661 €	8,268 €

Table 5: Comparison of BEL values under male and female mortality tables, stochastic vs deterministic projection.

Switching from male to female mortality tables results in a decrease in the BEL in both projection approaches, as lower mortality rates imply fewer death benefit payments. However, in both cases (different insured age and male Vs female analysis) the magnitude of the reduction is significantly larger in the stochastic case than in the deterministic one. This difference is driven by the death benefit guarantee $\max(C_0, F)$. As discussed in Section 4.1, in the deterministic projection the fund rarely falls below C_0 , so each death event costs approximately F . In the stochastic case instead, adverse market paths can push the fund below C_0 , increasing the average cost per death event. Switching to female mortality tables therefore has a larger impact in the stochastic projection, where each avoided death saves the full guaranteed amount C_0 rather than just F .

6 Python Code

```
import numpy as np
import pandas as pd

"""
Initialize the values of the Assignment
"""
# Set the seed for reproducibility, number of simulations:
np.random.seed(42)
N_SIM = 1000000

# Initial values
N_ASSICURATI = 100
PREMIO_UNICO = 100000
ORIZZONTE = 50

# Composition and volatility of the fund
W_EQ = 0.80; VOL_EQ = 0.20
W_PR = 0.20; VOL_PR = 0.10

# Fees and costs
RD_FEE = 0.0220; COMM_FEE = 0.0140
SPESA_FISSA = 50; INFLAZIONE = 0.02
LAPSE_BASE = 0.15; PENALE = 20

# symmetric adjustment
SA = 0.079

"""
Import the data from EIOPA (rates) and survival probabilities
"""
# Aggiunto il parametro opzionale "dec" per gestire virgole o punti
# (aggiunto per la prima domanda aperta dell'assignment)
def load_rates(filename, dec=','):
    df = pd.read_csv(filename, sep=';', decimal=dec)
    df.columns = df.columns.str.strip()
    return df.iloc[:, 1].values[:ORIZZONTE+1]

spot_base = load_rates('eiopa_rates.csv')
spot_up = load_rates('eiopa_rates_up.csv')
spot_down = load_rates('eiopa_rates_down.csv')

df_mort = pd.read_csv('tavola_SI2024.csv', sep=';')
df_mort.columns = df_mort.columns.str.strip()
qx_base = df_mort['qx'].values[:ORIZZONTE]

"""
Generating the scenarios via Antithetic Simulations
"""
```

```

N_HALF = N_SIM // 2

z_eq_half = np.random.standard_normal((N_HALF, ORIZZONTE))
z_pr_half = np.random.standard_normal((N_HALF, ORIZZONTE))

z_eq_global = np.concatenate((z_eq_half, -z_eq_half), axis=0)
z_pr_global = np.concatenate((z_pr_half, -z_pr_half), axis=0)

"""
BEL computation function:
"""
def calcola_BEL(spot_curve, qx_curve, lapse_rate, z_eq, z_pr,
    lapse_mass=0.0, spesa_mult=1.0, inf_add=0.0, shock_eq=0.0,
    shock_pr=0.0, deterministic=False):

    fwd = np.zeros(ORIZZONTE)
    for t in range(ORIZZONTE):
        fwd[t] = np.log((1 + spot_curve[t+1])**t / (1 +
spot_curve[t])**t)

    # DETERMINISTIC PART
    if deterministic:
        fund_det = np.zeros(ORIZZONTE + 1)
        fund_det[0] = PREMIO_UNICO * (1 - (W_EQ*shock_eq + W_PR*
shock_pr))
        comm_det = np.zeros(ORIZZONTE)

        for t in range(ORIZZONTE):
            fondo_lordo = fund_det[t] * np.exp(fwd[t])
            comm_det[t] = fondo_lordo * COMM_FEE
            fund_det[t+1] = fondo_lordo * (1 - RD_FEE)

        n_vivi = np.zeros(ORIZZONTE + 1); n_vivi[0] = N_ASSICURATI
        morti = np.zeros(ORIZZONTE); riscatti = np.zeros(ORIZZONTE)

        for t in range(ORIZZONTE):
            morti[t] = n_vivi[t] * qx_curve[t]
            if t == 0 and lapse_mass > 0:
                riscatti_massivi = (n_vivi[t] - morti[t]) *
lapse_mass
            else:
                riscatti_massivi = 0

            if t == ORIZZONTE - 1:
                riscatti[t] = n_vivi[t] - morti[t]
            else:
                riscatti[t] = (n_vivi[t] - morti[t] -
riscatti_massivi) * lapse_rate + riscatti_massivi
            n_vivi[t+1] = n_vivi[t] - morti[t] - riscatti[t]

        bel_det = 0.0

```

```

    # Contatori per la scomposizione deterministica
    bel_morte_det = 0.0; bel_riscatto_det = 0.0; bel_spese_det
= 0.0; bel_comm_det = 0.0

    for t in range(ORIZZONTE):
        v_fondo = fund_det[t+1]
        ben_morte      = morti[t] * max(PREMIO_UNICO, v_fondo)
        penale_t       = 0 if t == ORIZZONTE - 1 else PENALE
        ben_riscatto    = riscatti[t] * max(0, v_fondo - penale_t)
        spese           = n_vivi[t] * (SPESA_FISSA * spesa_mult) *
(1 + (INFLAZIONE + inf_add))**(t+1)
        comm_pagate    = n_vivi[t] * comm_det[t]

        df_det = 1 / (1 + spot_curve[t+1])**t
        cf = ben_morte + ben_riscatto + spese + comm_pagate

        bel_morte_det += ben_morte * df_det
        bel_riscatto_det += ben_riscatto * df_det
        bel_spese_det += spese * df_det
        bel_comm_det += comm_pagate * df_det

        bel_det += cf * df_det

    dettagli_det = {'Morte': bel_morte_det, 'Riscatto':
bel_riscatto_det, 'Spese': bel_spese_det, 'Commissioni':
bel_comm_det}
    return bel_det, 0, dettagli_det

# STOCHASTIC PART
eq_paths = np.zeros((N_SIM, ORIZZONTE + 1)); eq_paths[:, 0] =
PREMIO_UNICO * (1 - shock_eq)
pr_paths = np.zeros((N_SIM, ORIZZONTE + 1)); pr_paths[:, 0] =
PREMIO_UNICO * (1 - shock_pr)
fund_paths = np.zeros((N_SIM, ORIZZONTE + 1)); fund_paths[:, 0]
= PREMIO_UNICO * (1 - (W_EQ*shock_eq + W_PR*shock_pr))

rd_matrix = np.zeros((N_SIM, ORIZZONTE))
comm_matrix = np.zeros((N_SIM, ORIZZONTE))

for t in range(ORIZZONTE):
    eq_lordo = eq_paths[:, t] * np.exp((fwd[t] - 0.5*VOL_EQ**2)
+ VOL_EQ*z_eq[:, t])
    pr_lordo = pr_paths[:, t] * np.exp((fwd[t] - 0.5*VOL_PR**2)
+ VOL_PR*z_pr[:, t])
    fondo_lordo = W_EQ * eq_lordo + W_PR * pr_lordo

    rd_t = fondo_lordo * RD_FEE
    comm_t = fondo_lordo * COMM_FEE

    rd_matrix[:, t] = rd_t
    comm_matrix[:, t] = comm_t

```

```

    fund_paths[:, t+1] = fondo_lordo - rd_t
    eq_paths[:, t+1] = eq_lordo * (1 - RD_FEE)
    pr_paths[:, t+1] = pr_lordo * (1 - RD_FEE)

n_vivi = np.zeros(ORIZZONTE + 1); n_vivi[0] = N_ASSICURATI
morti = np.zeros(ORIZZONTE); riscatti = np.zeros(ORIZZONTE)

for t in range(ORIZZONTE):
    morti[t] = n_vivi[t] * qx_curve[t]

    if t == 0 and lapse_mass > 0:
        riscatti_massivi = (n_vivi[t] - morti[t]) * lapse_mass
    else:
        riscatti_massivi = 0
    if t == ORIZZONTE - 1:
        riscatti[t] = n_vivi[t] - morti[t]
    else:
        riscatti[t] = (n_vivi[t] - morti[t] - riscatti_massivi)
* lapse_rate + riscatti_massivi

    n_vivi[t+1] = n_vivi[t] - morti[t] - riscatti[t]

# BEL calculation with decomposition
bel_scenari = np.zeros(N_SIM)
dur_scenari = np.zeros(N_SIM)

bel_morte_scenari = np.zeros(N_SIM)
bel_riscatto_scenari = np.zeros(N_SIM)
bel_spese_scenari = np.zeros(N_SIM)
bel_comm_scenari = np.zeros(N_SIM)

for t in range(ORIZZONTE):
    v_fondo = fund_paths[:, t+1]

    ben_morte = morti[t] * np.maximum(PREMIO_UNICO, v_fondo)
    penale_t = 0 if t == ORIZZONTE - 1 else PENALE
    ben_riscatto = riscatti[t] * np.maximum(0, v_fondo -
penale_t)

    spese = n_vivi[t] * (SPESA_FISSA * spesa_mult) * (1 + (
INFLAZIONE + inf_add))**(t+1)
    comm_pagate = n_vivi[t] * comm_matrix[:, t]

    df = 1 / (1 + spot_curve[t+1])**t

    bel_morte_scenari += ben_morte * df
    bel_riscatto_scenari += ben_riscatto * df
    bel_spese_scenari += spese * df
    bel_comm_scenari += comm_pagate * df

```

```

        cf_netto = ben_morte + ben_riscatto + spese + comm_pagate
        cf_attualizzato = cf_netto * df
        bel_scenari += cf_attualizzato
        dur_scenari += (t + 1) * cf_attualizzato

    dur_media = np.mean(dur_scenari / bel_scenari)

    dettagli = {
        'Morte': np.mean(bel_morte_scenari),
        'Riscatto': np.mean(bel_riscatto_scenari),
        'Spese': np.mean(bel_spese_scenari),
        'Commissioni': np.mean(bel_comm_scenari)
    }

    return np.mean(bel_scenari), dur_media, dettagli

# Deterministic comparison
BEL_det, _, det_det = calcola_BEL(spot_base, qx_base, LAPSE_BASE,
    z_eq_global, z_pr_global, deterministic=True)

"""
Solvency Capital Requirement (SCR) computation
"""
results = {}

# BASE PART
MVA_0 = N_ASSICURATI * PREMIO_UNICO
BEL_base, dur_base, det_base = calcola_BEL(spot_base, qx_base,
    LAPSE_BASE, z_eq_global, z_pr_global)
BOF_base = MVA_0 - BEL_base
results['BASE'] = (MVA_0, BEL_base, BOF_base, dur_base, det_base)

# MARKET RISK PART
# Equity
shock_eq = 0.39 + SA
BEL_eq, dur_eq, det_eq = calcola_BEL(spot_base, qx_base, LAPSE_BASE
    , z_eq_global, z_pr_global, shock_eq=shock_eq)
MVA_eq = MVA_0 * (1 - W_EQ * shock_eq)
BOF_eq = MVA_eq - BEL_eq
SCR_Equity = max(0, BOF_base - BOF_eq)
results['EQUITY'] = (MVA_eq, BEL_eq, BOF_eq, dur_eq, det_eq)

# Property
shock_pr = 0.25
BEL_pr, dur_pr, det_pr = calcola_BEL(spot_base, qx_base, LAPSE_BASE
    , z_eq_global, z_pr_global, shock_pr=shock_pr)
MVA_pr = MVA_0 * (1 - W_PR * shock_pr)
BOF_pr = MVA_pr - BEL_pr
SCR_Property = max(0, BOF_base - BOF_pr)
results['PROPERTY'] = (MVA_pr, BEL_pr, BOF_pr, dur_pr, det_pr)

```

```

# Interest rate
BEL_int_up, dur_int_up, det_int_up = calcola_BEL(spot_up, qx_base,
    LAPSE_BASE, z_eq_global, z_pr_global)
BOF_int_up = MVA_0 - BEL_int_up
SCR_Int_up = max(0, BOF_base - BOF_int_up)
results['IR_UP'] = (MVA_0, BEL_int_up, BOF_int_up, dur_int_up,
    det_int_up)

BEL_int_dw, dur_int_dw, det_int_dw = calcola_BEL(spot_down, qx_base
    , LAPSE_BASE, z_eq_global, z_pr_global)
BOF_int_dw = MVA_0 - BEL_int_dw
SCR_Int_dw = max(0, BOF_base - BOF_int_dw)
results['IR_DOWN'] = (MVA_0, BEL_int_dw, BOF_int_dw, dur_int_dw,
    det_int_dw)

SCR_Interest = max(SCR_Int_up, SCR_Int_dw)
is_IR_down_worst = SCR_Int_dw > SCR_Int_up

# LIFE UNDERWRITING RISK PART
# Mortality
qx_mort = np.minimum(1.0, qx_base * 1.15)
BEL_mort, dur_mort, det_mort = calcola_BEL(spot_base, qx_mort,
    LAPSE_BASE, z_eq_global, z_pr_global)
BOF_mort = MVA_0 - BEL_mort
SCR_Mortality = max(0, BOF_base - BOF_mort)
results['MORTALITY'] = (MVA_0, BEL_mort, BOF_mort, dur_mort,
    det_mort)

# Lapse
lapse_up = min(1.0, LAPSE_BASE * 1.50)
BEL_lapse_up, dur_lapse_up, det_lapse_up = calcola_BEL(spot_base,
    qx_base, lapse_up, z_eq_global, z_pr_global)
BOF_lapse_up = MVA_0 - BEL_lapse_up
SCR_Lapse_up = max(0, BOF_base - BOF_lapse_up)
results['LAPSE_UP'] = (MVA_0, BEL_lapse_up, BOF_lapse_up,
    dur_lapse_up, det_lapse_up)

lapse_dw = max(0.5 * LAPSE_BASE, LAPSE_BASE - 0.20)
BEL_lapse_dw, dur_lapse_dw, det_lapse_dw = calcola_BEL(spot_base,
    qx_base, lapse_dw, z_eq_global, z_pr_global)
BOF_lapse_dw = MVA_0 - BEL_lapse_dw
SCR_Lapse_dw = max(0, BOF_base - BOF_lapse_dw)
results['LAPSE_DOWN'] = (MVA_0, BEL_lapse_dw, BOF_lapse_dw,
    dur_lapse_dw, det_lapse_dw)

# Mass lapse
BEL_lapse_mass, dur_lapse_mass, det_lapse_mass = calcola_BEL(
    spot_base, qx_base, LAPSE_BASE, z_eq_global, z_pr_global,
    lapse_mass=0.40)
BOF_lapse_mass = MVA_0 - BEL_lapse_mass
SCR_Lapse_mass = max(0, BOF_base - BOF_lapse_mass)

```

```

results['LAPSE_MASS'] = (MVA_0, BEL_lapse_mass, BOF_lapse_mass,
    dur_lapse_mass, det_lapse_mass)

SCR_Lapse = max(SCR_Lapse_up, SCR_Lapse_dw, SCR_Lapse_mass)

# Life cat
qx_cat = qx_base.copy()
qx_cat[0] = min(1.0, qx_cat[0] + 0.0015)
BEL_cat, dur_cat, det_cat = calcola_BEL(spot_base, qx_cat,
    LAPSE_BASE, z_eq_global, z_pr_global)
BOF_cat = MVA_0 - BEL_cat
SCR_Cat = max(0, BOF_base - BOF_cat)
results['CATASTROPHY'] = (MVA_0, BEL_cat, BOF_cat, dur_cat, det_cat
    )

# Expense
BEL_exp, dur_exp, det_exp = calcola_BEL(spot_base, qx_base,
    LAPSE_BASE, z_eq_global, z_pr_global, spesa_mult=1.10, inf_add
    =0.01)
BOF_exp = MVA_0 - BEL_exp
SCR_Expense = max(0, BOF_base - BOF_exp)
results['EXPENSES'] = (MVA_0, BEL_exp, BOF_exp, dur_exp, det_exp)

"""
BSCR aggregation:
"""
A = 0.50 if is_IR_down_worst else 0.00

Var_Market = (SCR_Equity**2 + SCR_Property**2 + SCR_Interest**2
    + 2 * 0.75 * SCR_Equity * SCR_Property
    + 2 * A * SCR_Interest * SCR_Equity
    + 2 * A * SCR_Interest * SCR_Property)
SCR_Market = np.sqrt(Var_Market)

Var_Life = (SCR_Mortality**2 + SCR_Lapse**2 + SCR_Expense**2 +
    SCR_Cat**2
    + 2 * 0.25 * SCR_Mortality * SCR_Expense
    + 2 * 0.25 * SCR_Mortality * SCR_Cat
    + 2 * 0.50 * SCR_Lapse * SCR_Expense
    + 2 * 0.25 * SCR_Lapse * SCR_Cat
    + 2 * 0.25 * SCR_Expense * SCR_Cat)
SCR_Life = np.sqrt(Var_Life)

BSCR = np.sqrt(SCR_Market**2 + SCR_Life**2 + 2 * 0.25 * SCR_Market
    * SCR_Life)

"""
Print final results
"""
print("BASE DELIVERABLES & SCR")
print(f"BEL deterministic: {BEL_det:,.0f} \n")

```

```

print(f"BEL Base:           {BEL_base:,.0f}      ")
print(f"MVA Base:           {MVA_0:,.0f}        ")
print(f"BOF Base:             {BOF_base:,.0f}      \n")

print(f"SCR Interest:        {SCR_Interest:,.0f}    ")
print(f"SCR Equity:            {SCR_Equity:,.0f}      ")
print(f"SCR Property:          {SCR_Property:,.0f}    ")
print(f"SCR Mortality:         {SCR_Mortality:,.0f}    ")
print(f"SCR Lapse:             {SCR_Lapse:,.0f}       ")
print(f"SCR Life Cat:          {SCR_Cat:,.0f}         ")
print(f"SCR Expense:           {SCR_Expense:,.0f}     \n")

print(f"BSCR FINALE:         {BSCR:,.0f}          \n")

"""
Martingale test
"""
def test_martingala_equity(spot_curve, z_eq):
    print("\n--- TEST DI MARTINGALA (SOLO EQUITY) ---")

    fwd = np.zeros(ORIZZONTE)
    for t in range(ORIZZONTE):
        fwd[t] = np.log((1 + spot_curve[t+1])**t / (1 +
spot_curve[t])**t)

    # Initial value only of the equity, since we want to do the
test only on equity
    EQ_0 = PREMIO_UNICO * N_ASSICURATI * W_EQ

    eq_pure = np.zeros(N_SIM); eq_pure[:] = EQ_0

    medie_attualizzate = np.zeros(ORIZZONTE)

    for t in range(ORIZZONTE):
        eq_pure = eq_pure * np.exp((fwd[t] - 0.5*VOL_EQ**2) +
VOL_EQ*z_eq[:, t])

        media_trasversale_t = np.mean(eq_pure)
        medie_attualizzate[t] = media_trasversale_t / ((1 +
spot_curve[t+1])**t)

    media_finale = np.mean(medie_attualizzate)
    std_dev_finale = np.std(medie_attualizzate)

    print(f"Equity Iniziale (Teorico): {EQ_0:,.2f}      ")
    print(f"Media sui 50 anni ( ):          {media_finale:,.2f}      ")
    print(f"Deviazione Std ( ):              {std_dev_finale:,.2f}      ")

test_martingala_equity(spot_base, z_eq_global)

```

```

print("\nFinal datas")
print(f"{'Scenario':<12} | {'MVA':>12} | {'BEL':>12} | {'d_MVA abs':>12} | {'d_BEL abs':>12} | {'d_MVA rel':>10} | {'d_BEL rel':>10} | {'BoF':>12} | {'d_BoF':>12} | {'dur_L':>6}")
print("-" * 135)

ordine_stampa = ['BASE', 'IR_UP', 'IR_DOWN', 'EQUITY', 'PROPERTY', 'MORTALITY', 'EXPENSES', 'CATASTROPHY', 'LAPSE_UP', 'LAPSE_DOWN', 'LAPSE_MASS']

for scenario in ordine_stampa:
    mva, bel, bof, dur, det = results[scenario]

    if scenario == 'BASE':
        d_mva_abs = "-"
        d_bel_abs = "-"
        d_mva_rel = "-"
        d_bel_rel = "-"
        d_bof = "-"
    else:
        val_d_mva_abs = MVA_0 - mva
        val_d_bel_abs = bel - BEL_base
        val_d_mva_rel = (val_d_mva_abs / MVA_0) * 100
        val_d_bel_rel = (val_d_bel_abs / BEL_base) * 100
        val_d_bof = max(BOF_base - bof, 0)

        d_mva_abs = f"{val_d_mva_abs:,.2f}"
        d_bel_abs = f"{val_d_bel_abs:,.2f}"
        d_mva_rel = f"{val_d_mva_rel:,.2f}%"
        d_bel_rel = f"{val_d_bel_rel:,.2f}%"
        d_bof = f"{val_d_bof:,.2f}"

    print(f"{scenario:<12} | {mva:>12,.2f} | {bel:>12,.2f} | {d_mva_abs:>12} | {d_bel_abs:>12} | {d_mva_rel:>10} | {d_bel_rel:>10} | {bof:>12,.2f} | {d_bof:>12} | {dur:>6.2f}")

print("-" * 135)
print(f"{'BSCR':<12} | {'':>12} {'':>12} {'':>12} {'':>12} {'':>10} {'':>10} {'':>12} {'BSCR:>12,.2f} | {'':>6}")

# SCOMPOSIZIONE DEL BEL

print("\n" + "="*80)
print(f"{'SCOMPOSIZIONE DEL BEL (Best Estimate of Liabilities)':~80}")
print("="*80)
print(f"{'Scenario':<12} | {'Morte ( )':>13} | {'Riscatto ( )':>13} | {'Spese ( )':>11} | {'Comm. ( )':>11} | {'TOTALE BEL':>13}")

```

```
print("-" * 80)

for scenario in ordine_stampa:
    mva, bel, bof, dur, det = results[scenario]

    if scenario == 'BASE':
        print("-" * 80)

        morte = f"{det['Morte']:,.0f}"
        riscatto = f"{det['Riscatto']:,.0f}"
        spese = f"{det['Spese']:,.0f}"
        comm = f"{det['Commissioni']:,.0f}"
        tot_bel = f"{bel:,.0f}"

        print(f"{scenario:<12} | {morte:>13} | {riscatto:>13} | {spese:>11} | {comm:>11} | {tot_bel:>13}")

        if scenario == 'BASE':
            print("-" * 80)

print("="*80 + "\n")
```